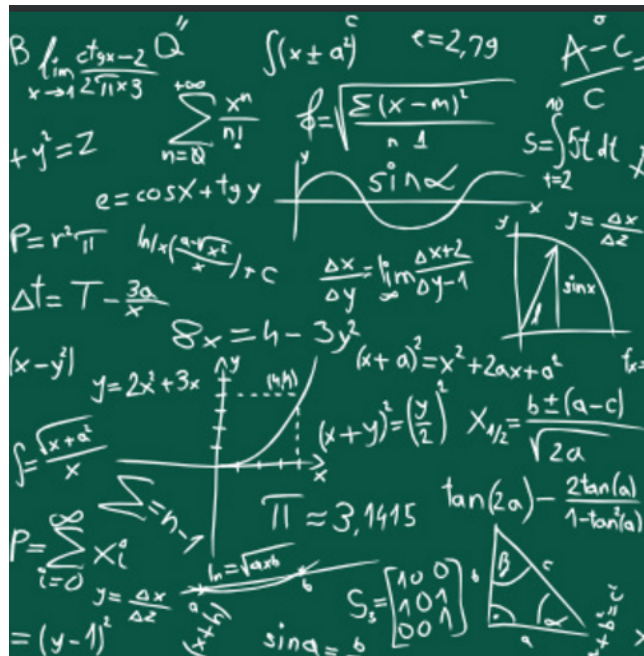


Monotone functions on an interval and invertibility



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- **Book:** "Analisi matematica 1", Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Monotone functions on an interval

- We now deal with monotone functions on an interval (and **not necessarily continuous**), and the next theorem (based on the completeness axiom of $\mathbb{R}$) can be seen as an extension to functions of the monotonicity theorem for sequences.

Theorem 1: Monotonicity theorem for functions

Let $f : (a, b) \rightarrow \mathbb{R}$ be a monotone function. Then for every $c \in (a, b)$ the right and left limits as $x \rightarrow c$ exist and are finite; at the two endpoints a, b the right limit (at a) and the left limit (at b) exist, possibly infinite.

Proof. Suppose f is increasing in (a, b) and let $c \in (a, b)$, that is, a point inside the interval.

We prove that the following limit exists and is finite:

$$\lim_{x \rightarrow c^-} f(x) = \sup \{f(x) : x \in (a, c)\}.$$

We set

$$\ell = \sup \{f(x) : x \in (a, c)\}$$

Note that ℓ exists and is finite by the supremum property, since $f(c)$ is an upper bound of the set $\{f(x) : x \in (a, c)\}$.

Hence we need to prove that

$$\lim_{x \rightarrow c^-} f(x) = \ell$$

So let $\{x_n\}$ be any sequence in (a, c) such that $x_n \rightarrow c$, and let us prove that $f(x_n) \rightarrow \ell$, i.e., that for every $\varepsilon > 0$ we eventually have

$$\ell - \varepsilon < f(x_n) < \ell + \varepsilon.$$

The second inequality is obvious, since $f(x_n) \leq \ell$ by definition of ℓ , because $x_n \in (a, c)$.

□

Proof. To prove the first one, we observe that, since $\ell - \varepsilon$ is less than ℓ , i.e., less than the least upper bound of $\{f(x) : x \in (a, c)\}$, it is not an upper bound of this set; therefore there exists a point

$$\tilde{x} \in (a, c) \text{ such that } f(\tilde{x}) > \ell - \varepsilon$$

Since f is increasing, it follows that

$$f(x) > \ell - \varepsilon, \forall x \in (\tilde{x}, c)$$

On the other hand, by hypothesis

$$x_n < c, \forall n \text{ and } x_n \rightarrow c \text{ as } n \rightarrow +\infty$$

therefore $x_n \in (\tilde{x}, c)$ eventually. Hence

$$f(x_n) \geq \ell - \varepsilon \quad (\text{eventually})$$

which is what we needed to prove.

Similarly, one can prove that the following limit exists and is finite:

$$\lim_{x \rightarrow c^+} f(x) = \inf \{f(x) : x \in (c, b)\}.$$

As for the limits at the two endpoints of the interval, we prove that the following limit exists:

$$\lim_{x \rightarrow b^-} f(x)$$

We set (as before)

$$\ell = \sup \{f(x) : x \in (a, b)\}$$

however, in this case ℓ could also be $+\infty$ (we cannot claim that $f(b)$ is an upper bound of the set, because the function is not defined at b). We therefore proceed by cases.

In the case $\ell < \infty$ the previous proof can be repeated. □

Proof. If instead $\ell = \infty$, the argument changes. From the hypothesis

$$\sup \{f(x) : x \in (a, b)\} = +\infty$$

it follows that for every $M > 0$ there exists $\tilde{x} \in (a, b)$ such that $f(\tilde{x}) > M$.

By the monotonicity of f , then

$$f(x) > M, \quad \forall x \in (\tilde{x}, b)$$

hence, taking any sequence $x_n \rightarrow b$, we have $x_n \in (\tilde{x}, b)$ eventually, and therefore

$$f(x_n) > M, \quad \text{eventually,}$$

therefore

$$\lim_{n \rightarrow +\infty} f(x_n) = +\infty$$

In the same way one proves that the following limit exists:

$$\lim_{x \rightarrow a^+} f(x)$$

□

A consequence of the monotonicity theorem is that if a function is monotone on an interval (a, b) , its possible points of discontinuity in (a, b) are necessarily jump discontinuities, except for the endpoints a, b , where there may also be a vertical asymptote.

2 Continuity and invertibility

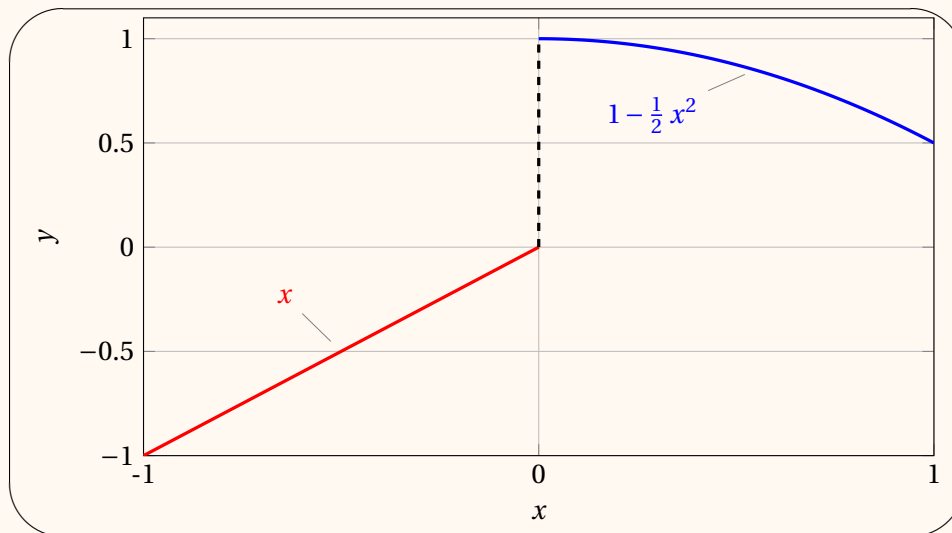
- We have seen the theorem stating that if a generic function with domain D is strictly monotone, then it is invertible.
- We also know that the converse is not true in general: there exist functions that are invertible on an interval but not monotone.

Example 1: Invertible but not monotone function

Consider for example

$$f(x) = \begin{cases} 1 - \frac{1}{2}x^2 & \text{if } 0 < x \leq 1, \\ x & \text{if } x \leq 0 \end{cases}$$

its graph is:



This function satisfies the invertibility condition, which requires the graph of f to be intersected at most at one point by every line parallel to the x -axis, but it is not monotone.

- If we add the hypothesis of continuity and require the domain to be an interval, being strictly monotone becomes a necessary and sufficient condition for invertibility, as stated by the following theorem.

Theorem 2: Invertibility of monotone and continuous functions

Let $f : I \rightarrow \mathbb{R}$ be a function defined on an interval I .

If the function f is continuous, then it is invertible on the interval I if and only if it is strictly monotone.

In this case its inverse function is also strictly monotone and continuous.

Proof. We already know that if f is strictly monotone, it is invertible (regardless of the hypotheses that f is continuous and that it is defined on an interval).

We show that the converse holds, i.e., that if f is continuous and invertible, then it is strictly monotone.

Suppose by contradiction that the function is not strictly monotone; then there exist three points

$$x_1 < x_2 < x_3 \text{ (in the interval } I \text{)}$$

such that

$$f(x_1) < f(x_2) \text{ and } f(x_2) > f(x_3)$$

or such that

$$f(x_1) > f(x_2) \text{ and } f(x_2) < f(x_3).$$

Suppose the first of the two alternatives holds (the other case is handled similarly). Let us compare the values $f(x_1)$ and $f(x_3)$; they cannot be equal because f is invertible by hypothesis, hence

$$f(x_1) < f(x_3) \text{ or } f(x_1) > f(x_3).$$

Again, suppose the first of the two alternatives holds (the other case is handled similarly). So we know that:

$$x_1 < x_2 < x_3 \text{ and } f(x_1) < f(x_3) < f(x_2).$$

Since f is continuous, by the intermediate value theorem there exists

$$x_0 \in (x_1, x_2) \text{ such that } f(x_0) = f(x_3).$$

Since $x_0 \neq x_3$ (because $x_1 < x_2 < x_3$), it follows that f cannot be invertible, a **contradiction**. This proves the first part of the theorem.

Now let f be a continuous, strictly monotone and hence invertible function on I , and let g be its inverse function, which is also strictly monotone and invertible. We prove that g is continuous.

- By what we observed after the monotonicity theorem, the strictly monotone function g is either continuous or has jump discontinuity points.
- In the latter case the image of g is not an interval (but the union of at least two disjoint intervals), which is a **contradiction** because this image is I . Hence g is continuous.

□

- Note that in the proof of the previous theorem we used both the intermediate value theorem and the monotonicity theorem for functions. Moreover, we implicitly used the completeness axiom of $\mathbb{R}$.
- The theorem just proved means, in particular, that:

a continuous and invertible function on an interval has a continuous inverse function.

- This fact completes the proof of the continuity theorem for elementary functions:
 1. the continuity of the function a^x implies the continuity of the function $\log_a x$.
 2. the continuity of the functions $\sin x$, $\cos x$ and $\tan x$ implies the continuity of the functions $\arcsin x$, $\arccos x$ and $\arctan x$.